

Lightweight Mutual Authentication between IoT Devices using One-way Accumulator

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Abstract—The Internet of Things (IoT) is integral to modern life, connecting diverse devices that collect and transfer data to central servers or the cloud. IoT devices are resource-constrained in nature and not equipped with adequate in-built security. This raises significant concerns regarding data security and integrity. Thus, secure and lightweight authentication schemes are of utmost need. Existing lightweight authentication schemes often depend on persistent connectivity with a central server and yield significant computational and communication overhead during authentication processes. To address the above issues, this paper proposes a lightweight mutual authentication scheme for IoT devices, leveraging a one-way accumulator while minimizing the dependency on a centralized entity. This cryptographic primitive enables the combination of multiple values into a single, difficult-to-reverse accumulated value, facilitating secure identity verification between nodes. To validate the proposed scheme, we conduct formal security analysis using the ProVerif tool. Additionally, we present a simulation-based comparative analysis using the Cooja IoT network simulator, highlighting the efficiency of our approach over existing benchmark authentication protocols in IoT environments.

Index Terms—IoT, Authentication, One-way Accumulator, Security, Lightweight Protocol

I. MOTIVATION

The Internet of Things (IoT) has integrated billions of devices, interconnecting networks [1] across sectors such as healthcare, transportation, and energy [2]. Although mutual authentication plays a pivotal role in safeguarding IoT nodes from security vulnerabilities, existing authentication schemes often require uninterrupted connectivity to a centralized server [3], [4], [5], or suffer from high computational overhead [6], [7]. Thus, these schemes create a bottleneck and a single point of failure, rendering them unsuitable for resource-constrained IoT networks. These problems are addressed in this work, where we propose a lightweight mutual authentication scheme using an RSA-based one-way accumulator. This scheme aims to minimize dependency on a central authority while maintaining security requirements. It also supports lightweight cryptographic operations (hashing and XOR) during frequent authentication.

II. CONTRIBUTIONS

The main contributions of this paper are as follows:

- A lightweight mutual authentication scheme is proposed using an RSA-based one-way accumulator for

IoT devices, enabling secure authentication in resource-constrained environments with reduced reliance on a central authority during the authentication phase.

- An extensive formal security analysis is performed, indicating the validity of the proposed lightweight scheme to resist the leakage of authentication secrets and to establish mutual authentication.
- A performance evaluation using the Cooja simulator and a theoretical framework is performed, measuring both energy consumption and CPU time to validate the efficiency of the proposed scheme in resource-constrained IoT networks.

III. RELATED WORK

Existing schemes such as [3] and [5] demonstrate heavy cryptographic operations and a persistent dependence on a Central Authority (CA), leading to a single point of failure and high resource overhead. Even mutual authentication schemes like [7], [8], and [9] rely on heavy operations or have higher communication overhead, making them unsuitable for IoT. The proposed scheme addresses these limitations through low resource overhead and CA-free authentication.

TABLE I: Features comparison of authentication schemes.

Scheme	Crypto Ops	Comp. OH	Mutual Auth.	Enc/Dec	Comm. OH	CA-Indep.
[5]	Low	Medium	X	✓	Medium	X
[3]	High	High	X	✓	High	X
[8]	Medium	High	✓	✓	Medium	X
[9]	High	Medium	✓	✓	Low	X
[7]	Medium	Low	✓	X	Medium	✓
Proposed Scheme	Low	Low	✓	X	Low	✓

Legend: ✓ = Achieved, X = Not Achieved.

Crypto Ops: Cryptographic operations in registration/authentication — Low (1–19), Medium (20–39), High (≥ 40).

Comp. OH: Computational Overhead — Low (Hash/XOR), Medium (Symmetric/light ECC), High (Heavy ECC/Pairings).

Enc/Dec: Uses heavy encryption in authentication phase.

Comm. OH: Message exchanges — Low (1–2), Medium (3–4), High (≥ 5).

CA-Indep.: End devices authenticate without CA/TTP post-registration.

IV. PROPOSED LIGHTWEIGHT MUTUAL AUTHENTICATION SCHEME

In the context of resource-constrained environments such as IoT, RSA-based accumulators offer a promising balance between security and performance, motivating their incorporation into our lightweight mutual authentication protocol

TABLE II: Notations used in the proposed scheme.

Notation	Description
CA	Central Authority
p, q	Large prime numbers
N	RSA modulus
s	Accumulator base (seed)
$Node_i$	Node i
x_i	Secret value unique to $Node_i$
k	Total number of nodes
w_i	Witness for $Node_i$
Z_{global}	Accumulated value computed by CA
Z_A, Z_B	Accumulated value for $Node_A$ and $Node_B$
$H(\cdot)$	Cryptographic hash function
Z_A^H, Z_B^H	Hashed accumulated values
$Nonce_A, Nonce_B$	Nonces generated by $Node_A$ and $Node_B$
R_A, R_B	Random components of the nonces
t_A, t_B	Timestamp components of the nonces
Δ	Maximum acceptable time difference
ID_A, ID_B	Identifiers of $Node_A$ and $Node_B$
$K_A, K'_A, K_A^*, K_B, K'_B, K_B^*$	Intermediate cryptographic values
M_A, M_B	Masked messages of $Node_A$ and $Node_B$
$\parallel$	Concatenation operator
$\oplus$	Bitwise XOR operator
$==$	Equality check

A. Node Registration Phase

In the registration phase, every IoT node registers with a trusted CA to acquire cryptographic credentials for future independent mutual authentication. Figure 1 illustrates the overall workflow. This one time secure channel procedure with the CA provides each node with its witness (w_i , generated from its unique secret x_i) and the RSA modulus N . First, the

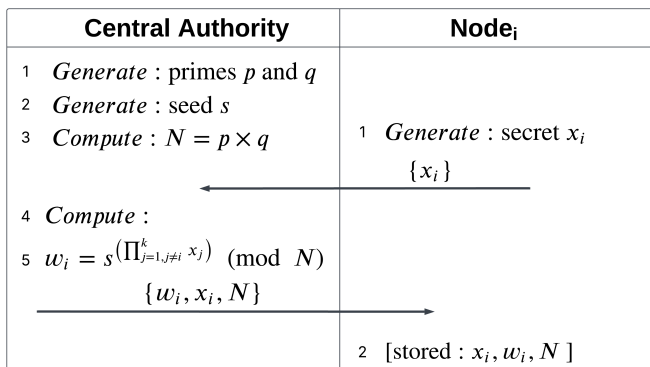


Fig. 1: Node Registration Phase with Central Authority (CA).

CA prepares the RSA modulus N (shown in step 3 of Figure 1) and the seed s as accumulator parameters. On the other hand, each IoT node ($Node_i$) generates its own unique secret (x_i) and sends it to the CA through a secure channel. The CA then computes w_i for each node ($Node_i$) by excluding its own secret (x_i) and securely returns (w_i, x_i, N) to the respective $Node_i$. Each node then stores the received credentials for future authentication.

B. Mutual Authentication Phase

Following successful registration with the CA, Node A and Node B can independently authenticate one another. Each one of them computes their respective accumulator values Z_A and Z_B , which match only if both are legitimate nodes. A two-message exchange with fresh nonces, as shown in Figure 2, secures the process over an insecure channel. Node A initiates the process by generating a nonce $Nonce_A$ and computing Z_A along with its hash Z_A^H . For the secure communication of Z_A^H , Node A derives a session-specific value K_A . This K_A is used to mask Z_A^H to produce M_A , preserving the confidentiality of Z_A^H . $(M_A, Nonce_A, ID_A)$ are then sent to Node B by Node A . Node B verifies the nonce freshness and computes Z_B . Then, Node B checks whether the recovered Z_A from Node A matches the computed Z_B , thereby authenticating A . After this, B returns $(M_B, Nonce_B, ID_B)$ to A , computed in the same way as A . Node A verifies the message following the same procedure and authenticates B .

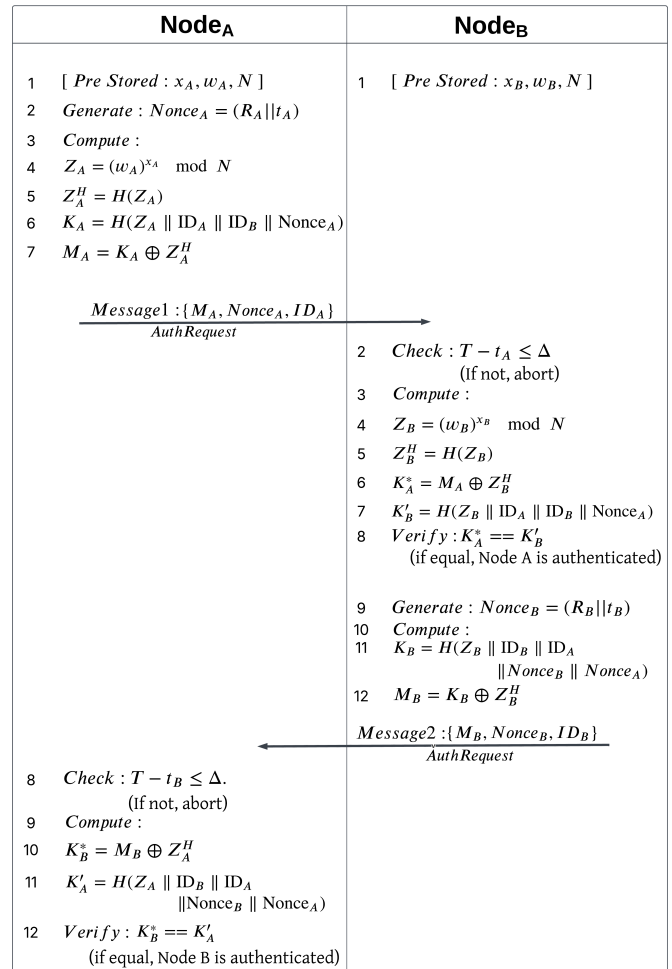


Fig. 2: Mutual authentication phase between Nodes A and B.

V. FORMAL SECURITY ANALYSIS USING PROVERIF

ProVerif offers formal verification for the proposed RSA-based accumulator scheme using Pi-calculus. All essential components, including communication channels, constants, and key cryptographic operations, are formally defined, representing the confidentiality aspect of the scheme. Registration and authentication phases are both modeled in this section.

In Figures 3 and 4, all queries related to the secret of Node i (x_i), the accumulator values of nodes A and B (Z_A , Z_B), and the session keys of nodes A and B (K_A , K_B) return *true*, ensuring their resistance to leakage.

In Figure 4, following event correspondence queries, $event(AuthenticatedA(id)) \Rightarrow event(AuthRequestbyA(id))$ and $event(AuthenticatedB(id)) \Rightarrow event(AuthRequestbyB(id))$ validate that both nodes A and B successfully authenticate each other.

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Verification summary:
Query not attacker(Secretx_i[]) is true.
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nihal@nihal-labPC:~/Documents$
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Fig. 3: ProVerif output showing secrecy of x_i .

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Verification summary:
Query not attacker(SecretZ_a[]) is true.
Query not attacker(Secretk_a[]) is true.
Query not attacker(SecretZ_b[]) is true.
Query not attacker(Secretk_b[]) is true.
Query event(AuthenticatedA(id)) ==> event(AuthRequestbyA(id)) is true.
Query event(AuthenticatedB(id)) ==> event(AuthRequestbyB(id)) is true.
-----
nihal@nihal-labPC:~/Documents$
```

Fig. 4: ProVerif output showing secrecy of intermediate values and authentication correspondence.

VI. PERFORMANCE ANALYSIS

A. Theoretical Analysis - Computational Cost

We measured the computational cost of key cryptographic operations like SHA-256 (T_h), modular exponentiation (T_{me}), random generation (T_r), ECC multiplication (T_{ecc}), AES (T_{aes}), and XOR (T_x), using the PyCrypto library on an Apple M1 (8-core).

1) Computational Cost in Mutual Authentication Phase

In the mutual authentication phase (Figure 2), both Node A and Node B execute one random generation, one modular exponentiation, three hash, and two XOR operations, leading to an overall session cost of $2T_r + 2T_{me} + 6T_h + 4T_x$. As demonstrated in Table III, our proposed scheme exhibits the lowest computational cost (0.0648 ms) among all the compared schemes. This efficiency results from the integration of

TABLE III: Comparison of Computational Costs: Mutual Authentication Phase

Scheme	Operations	Cost (ms)
[7]	$12T_h + 12T_x + 7T_r$	0.1670
[10]	$19T_h + 6T_x + 5T_r$	0.1800
[11]	$16T_h + 11T_x + 2T_r$	0.1850
[8]	$11T_h + 2T_x + 2T_r + 10T_{ecc}$	6.9100
[9]	$27T_h + 34T_x + 3T_r + 3T_{aes}$	0.4390
Proposed Scheme	$2T_r + 2T_{me} + 6T_h + 4T_x$	0.0648

Benchmark (avg. of 100 iterations):

$T_r = 0.0010$ ms (Random Generation), $T_h = 0.0073$ ms (SHA-256), $T_x = 0.0006$ ms (XOR), $T_{me} = 0.0083$ ms (Modular Exp.), $T_{aes} = 0.0117$ ms (AES), $T_{ecc} = 0.6816$ ms (ECC Mult.).

Platform: Apple M1, 8-core CPU, macOS 15.

lightweight hash and XOR operations while limiting heavy computation to only one modular exponentiation.

B. Theoretical Analysis - Communication Cost

The communication cost is primarily determined by the number of messages exchanged and their sizes, assuming a uniform bit-length framework. Each node identity and timestamp are assumed to be 32 bits. Random nonces, SHA-256 digests, and AES ciphertext blocks are considered to be 256 bits. Elliptic curve points are represented in 512 bits, along with the RSA modulus and accumulator witnesses as 1024 bits.

1) Communication Cost in Mutual Authentication Phase

The mutual authentication phase requires two message exchanges between nodes A and B , each consisting of a 32-bit ID, a 288-bit nonce, and a 256-bit masked value. Thus, the total size per exchange is 576 bits, leading to an aggregate communication overhead of 1152 bits per session.

TABLE IV: Comparison of Communication Costs: Mutual Authentication Phase

Scheme	Number of Messages	Cost (bits)
[7]	3	2176
[10]	4	2176
[11]	3	2048
[8]	3	2304
[9]	2	1344
Proposed Scheme	2	1152

All schemes assume: temporary IDs and timestamps = 32 bits; random nonces, hash digests, and symmetric ciphertexts = 256 bits; ECC points (P-256) = 512 bits; RSA moduli or witnesses = 1024 bits unless specified otherwise.

As demonstrated in Table IV, our proposed scheme achieves a minimal communication cost of 1152 bits through a two-message exchange. It avoids bulky key transmissions by incorporating accumulator credentials. Thus, ensure low communication overhead and efficiency in a resource-constrained IoT environment.

C. Simulation-based Performance Comparison

We tested our authentication scheme in Cooja (Contiki-NG) using parameters listed in Table V. A 55-node RPL network is deployed randomly, with Node 1 as the Central Authority (CA) and the remaining 54 client nodes performing registration and mutual authentication via CoAP. Energy and CPU usage are calculated and compared with Interaction-Based Authentication (IBA) [7] leveraging Alkesh *et al.*'s formula [12]. Each experiment is executed 10 times for consistency.

TABLE V: Simulation Parameters

Parameter	Value
Operating System	Contiki-NG
Mote Type	Cooja Mote
Network Size	5, 15, 55 Nodes
Propagation Model	UDGM (Distance Loss)
Application Layer	CoAP
Network Layer Protocol	RPL Lite
MAC Layer Protocol	CSMA
Simulation Time	75 Seconds

1) Performance under One-time Mutual Authentication

The first set of experiments examines the overhead of one-time mutual authentication for node pairs. As shown in Figure 5, our scheme consistently outperforms IBA [7] with lesser energy consumption. In the 55-node network scenario (27 authenticating pairs), our scheme achieves a reduction of 47.1% in energy consumption. Figure 6 further indicates that the CPU time is reduced by 46.1% compared to IBA [7].

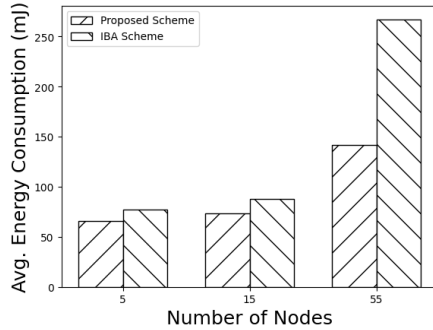


Fig. 5: Average energy consumption for a one-time authentication event per pair under varying network size

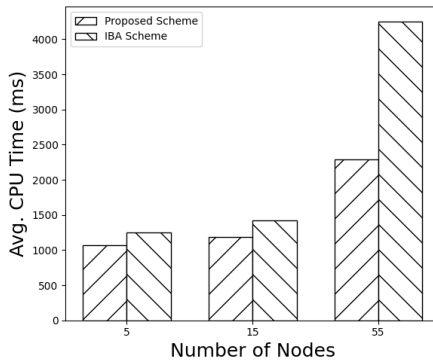


Fig. 6: Average CPU processing time for a one-time authentication event per pair under varying network size.

2) Performance under Multiple Mutual Authentications

Frequent authentication in IoT networks calls for performance analysis based on energy consumption and CPU time across multiple runs of the authentication phase. In a 55-node IoT network with 27 unique pairs, we performed repeated authentications 5, 10, and 15 times. Results in Figure 7 highlight that the proposed scheme outperforms IBA by 26.7% in

energy consumption per authentication at 15 authentications. In Figure 8, although CPU time per authentication grows with the number of authentication attempts, the proposed scheme showcases an advantage of 26.6% over IBA in the 15-authentication scenario.

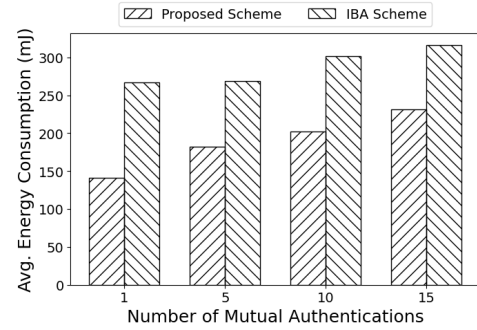


Fig. 7: Average Energy consumption per authentication under repeated mutual authentication events.

This is due to the optimized two-message exchange design in the proposed scheme, minimizing radio on-time. In addition, its reliance on lightweight cryptographic operations, compared to the repetitive interaction record processing of IBA [7], contributes to making the proposed scheme efficient.

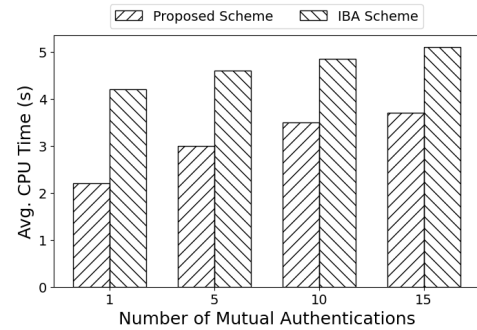


Fig. 8: Average CPU processing time per authentication under repeated mutual authentication events.

VII. CONCLUSION

In this paper, we present a lightweight mutual authentication scheme for an IoT environment that leverages an RSA-based accumulator. The scheme achieves secure and efficient bidirectional authentication between IoT nodes with minimal computational and communication overhead. They are only dependent on the CA during registration to perform authentication independently with the obtained credentials, eliminating the single point of failure. Both formal and experimental analyses validate its strong security and significantly lower resource overhead compared to other existing schemes.

VIII. ACKNOWLEDGEMENT

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